

Research Statement

My long-term research goal is to contribute methods and systems that realize ideal interfaces between people, artificial agents, and artifacts in virtual and physical environments—based on a deep understanding of the respective run-time interoperability requirements of these entities. This goal is based on my research over the past decade that builds on principles from ecological psychology and interoperability to yield results that apply to autonomous agents—including both, human and artificial agents. I situate this research in the Ubiquitous Computing field, and it has strong disciplinary interactions with the Human-Computer Interaction (HCI), Autonomous Agents and Multi-Agent Systems / Artificial Intelligence (AA), and Distributed Systems fields, interdisciplinary interactions with fields from Ecological Psychology (e.g., action perception) to Law (e.g., accountability), and applications from industrial robotics to health to social and environmental sustainability.

Across these fields, I have over the past six years built significant experience in the coordination and participation of large-scale disciplinary and interdisciplinary competitive research and innovation projects, including the SNSF *HyperAgents*¹ project (Co-PI; 2019-2024; AA+HCI; Fundamentals), the EU *IntellIoT*² research and innovation action (Co-PI; 2020-2024; AA; Robotics), the SNSF *FoodCoach*³ project (Partner; 2020-2024; HCI; Health and Nutrition), the Innosuisse *MR-MRO*⁴ project (Co-PI; 2021-2023; HCI; Robotics), the Innosuisse Flagship *WISER*⁵ (Co-PI; 2022-2026; AA+HCI; Environmental Sustainability), the SNSF *CoCoDa*⁶ project (Co-PI; 2025-2029; AA+HCI+Law; Social Sustainability), the TA-SWISS project *Sprachmodelle und Kompetenzverschiebungen* (Co-PI; 2025-2026; AA; Social Sustainability), the SNSF/Innosuisse Bridge Discovery project *Access to Justice*⁷ (Co-PI; 2026-2030; HCI+Law; Social Sustainability), and over a dozen industry- and foundation-financed projects.

Background and Current Research

Ph.D. Research. Together with colleagues and team members across several institutions, I have been researching interoperability since my Ph.D. This includes contributions to the Web of Things (WoT) to counter the fragmentation of the Internet of Things' (IoT) application layers (cf. [19, 34, 45, 46, 51]) and—striving for higher-level semantic interoperability—yielded automated planning approaches to compose services in WoT systems (cf. [33, 41, 40, 49, 50, 55]). Together with first links with affordance theory already in my Ph.D. thesis (cf. [39]), it formed part of the basis of the standardization of the World Wide Web Consortium's WoT Thing Descriptions (W3C WoT TDs) which are today the prime contender for application-layer interoperability in IoT systems in practice, with deployed products by several multinational engineering companies (e.g., the *Siemens Desigo* Building Management System series). For industry, a concrete realized advantage that my work contributed to is that the replacement of parts of an industrial system (such as sensors in a building) becomes much simpler due to a reduction of hard-coding to specific device or service APIs which massively reduces application-layer integration effort. Corresponding with this research, but in the HCI and Ubiquitous Computing fields, my Ph.D. also investigated the interaction of human users with WoT environments through the use of computer vision methods for the detection of [53], interaction with [54], and monitoring of [42, 47] interactable devices in IoT environments.

Interoperability in Technical Systems. After concluding my Ph.D., as part of my role as a *Senior Key Expert* for *Smart and Interacting Systems* with Siemens Corporate Technology in Berkeley, US, I continued to work on interoperability between technical systems [48] and user interfaces [2, 43, 67], and made several contributions to industrial products and published patents in the fields of technical and semantic interoperability

¹See <https://data.snf.ch/grants/grant/189474>

²See <https://cordis.europa.eu/project/id/957218>

³See <https://data.snf.ch/grants/grant/188402>

⁴<https://www.aramis.admin.ch/Grunddaten/?ProjectID=47839>

⁵<https://wiser-climate.com/>

⁶<https://data.snf.ch/grants/grant/10004598>

⁷Grant awarded in October 2025.

(e.g., interfaces for robot programming). My interest in the *integration* of the AA and HCI fields in a socio-technical perspective grew and, since then, I regularly publish on human-device interfacing and adaptation of device behavior (e.g., of run-time adaptive industrial robots [27]) in the HCI (e.g., [4, 6, 27, 32, 59, 69]) and Ubiquitous Computing (e.g., [5, 15, 68, 70]) fields, with additional consideration of explainability and increasing decision transparency in cyber-physical systems [31] as well as multi-modal context understanding [30]. Building on top of core architectural principles of the World Wide Web, I was successful in integrating this research with my group’s and others’ work in the Semantic Web and AA domains towards the conceptualization and definition of *Hypermedia Multi-Agent Systems (hMAS)* [10, 11, 66]. hMAS are a class of MAS that integrates the architectural properties of the World Wide Web and of MAS—towards scalable long-lived hybrid communities of people and artificial agents that are *dynamically open*, i.e., they support run-time interoperability with diverse agents. From a practice perspective, I am in this field today contributing to making artifacts in technical systems interoperable with arbitrary artificial agents, with partners from industry (cf. [62, 63]).

Interoperability for Artificial Agents and People. In the vocabulary of HCI, affordance theory (e.g., [9, 16, 58]) explains how animals might efficiently exploit interaction possibilities offered by their environment and by other agents and hence provides an eco-psychological conceptualization of interoperability. I argue that signifiers [58] should be conceptualized as environmental cues—for humans/animals *as well as* for artificial agents—that can be *intuitively and reliably* discovered and interpreted to provide guidance to agents about *what* are the possible behaviors in an environment and *how* these behaviors can be performed [81]. Humans experience such interoperability-by-design with (well-designed) signs, furniture, and, famously, door handles [58], and this idea established the field of User-Centered Design (UCD) [29]. To apply affordance theory to *artificial* agents, we however needed to capture it in specific engineering abstractions, and we proposed mathematical formalizations of the contextual exposure of signifiers to permit *run-time adaptation* of environmental signifiers [38, 81, 82]. This mechanism obtains information about the environment via discovered W3C WoT TDs (or other types of artifact manuals [82]) and combines this with information about the agent’s internal configuration (e.g., abilities and goals), which we formalize as part of *Agent Profiles* that are also discovered at run time [81, 82]. Agents are, in this way, able to receive guidance from the environment in a way that reflects the HATEOAS (hypermedia as the engine of application state) principle of the modern Web architecture [13]—and thereby may offload considerable computational effort (e.g., for model-building and intent selection) to the environment. A fitting analog is that this research brings “flow” [12] to the field of AI agents.

Dangers of Personalized Environmental Guidance. However, similar to a human who is enjoying such flow while surfing a trusted Web environment and simply clicks the most-highlighted hyperlink (think of *Accept all Cookies*), there is a trade-off between receiving (efficient) environmental guidance on the one hand and agent privacy risk and manipulation risk on the other. Social media and other engagement-oriented online platforms follow the same logic. With my group, I am therefore investigating methods to guarantee the *responsible* adaptation of signifiers for human users [69, 72] in applications such as diminished reality for healthier shopping [73] and the provisioning of opportunistic spatial reminders [17]. A major recent step was the definition and validation of a conceptual model for *Responsible Ubiquitous Personalization Systems* [71]—this paper also explains our expectation that personalization will, in the coming decades, not remain confined to online experiences in a 2D screen but rather extend to Ubiquitous Computing through Mixed-Reality- and IoT-based personalization, including personalization towards AI agents.

Future Research

My group’s recent and current research hence provides a conceptual bridge between hypermedia-driven affordance exploitation in HCI and methods for representing and reasoning about actions from AI research, and our results were disseminated to AA research (cf. [8, 28, 37, 38, 81, 82]) and practice (cf. [61, 62, 64]), including in the context of two Dagstuhl seminars that my group co-organized in 2021 (on *Autonomous Agents on the Web*,

Seminar #21072) and 2023 (on *Agents on the Web*, Seminar #23081). Through its grounding in affordance theory, our AI agents research is synchronized with our work in the HCI field, and we are currently working on the dissemination of the same concepts, but using a more suitable vernacular for the HCI and Ubiquitous Computing community. I expect that, in this way, we will be able to anchor my targeted conceptual integration *across artificial agents and humans* in the research community.

Cognitive Interoperability. Considering the increasing co-habitation of physical, virtual, and hybrid environments by humans and artificial agents—very timely given the recent and architecturally troubling introduction of Agent-based Web browsers such as Atlas⁸—we argue that UCD’s core design challenge to create efficient, effective, and highly satisfying user experiences remains highly topical, but in an extended form that includes *diverse artificial agents*. My goal is to enable humans and artificial agents to efficiently perceive contextually relevant information about interaction possibilities with their environment and each other, and thereby, together, reach their goals more efficiently and while requiring less design-time information that is supplied out-of-band. This requires an alignment of exchanged symbols not only on the syntactic and semantic levels, but to map them to each agent’s cognitive constructs: *cognitive interoperability* [57]. To this end, in [81, 82], we show how signifiers for the *same* environmental action possibility—actuating a robotic arm—can be exposed in *different* ways for Belief-Desire-Intention (BDI) [65] agents on the one hand and for agents that employ STRIPS-based reasoning [14] on the other. While signifiers for the BDI agents connect environmental action possibilities with the agent’s *desires*, STRIPS-oriented exposure relates them with *preconditions* and *effects* that are natural for AI planners like STRIPS. In a paper that is currently in review and that set in motion a new research stream in my group, we go one step further and show that we can expose signifiers that adjust to the cognitive constructs of BDI agents, Dyna [75] reinforcement learning agents, Soar [35] agents, and Large-Language-Model-based CoALA [74] agents, as well as to humans; with our adjustment mechanisms, the agents can achieve their goals together and while exploiting their respective non-functional profiles (e.g., trading-off CoALA’s high resource consumption with BDI’s relative inflexibility)—this hence realizes cross-cognitive interoperability for artificial agents, and we aim to generalize this to a model of cognitive interoperability over the next year.

Personalized Reality. Carrying over to Ubiquitous Computing and HCI research, my next steps are the investigation of mechanisms that increase the transparency of personalization mechanisms and enhance the agency that users—again, people as well as artificial agents—have over personalization. Based on [17, 69, 71, 72, 73], I seek to create and investigate prototype systems that implement *personalization sharing*: giving agents the ability to integrate elements of someone else’s personalized reality into their own, or even to swap their perceived reality with another person or artificial agent (cf. [69, 71, 72])—which is enabled by building on the same conceptual model for our AA- as well as our HCI-related research under a Ubiquitous Computing umbrella.

Socio-Technical Interoperability and Society. Summarizing, while interoperability has been studied in Computer Science (and other domains) for many years, I believe that our approach—thanks to its grounding in ecological psychology, AI, HCI, and Web research—has the potential to enable ad-hoc interoperability: across human and artificial agents in environments that are populated by diverse humans and diverse artificial agents; across physical, virtual, and hybrid environments; and including cognitive interoperability—all based on the same theoretical foundation. I believe that this holds the potential to provide insights on the structure of interoperability in socio-technical systems and yields the principled possibility to create interoperability-on-demand through personalized and contextualized run-time adaptive interfaces. Zooming out, our research in-between the AA and HCI fields has led me to the realization that UCD—a core rationale in HCI research—is incomplete. UCD today emphasizes that development processes need to put *the user* as the center of focus. However, this misses that user-centered design practices might lead to experiences that have large negative externalities *on society*. This is exemplified, on the micro level, through engagement-maximizing interfaces that

⁸See <https://openai.com/de-DE/index/introducing-chatgpt-atlas/>

afford societally undesired user behavior (e.g., doomscrolling); and on the macro level through experiences that are designed to trap users in emotionalized content filter bubbles; in other fields (e.g., casinos, medication, drugs) we regulate such consequences. In user and agent interfaces, these externalities need to be understood and measured (cf. our early work in [3]) while more and more language-model-powered AI agents are entering the environment, creating an ever more complex socio-technical landscape.

Integration at EPFL

In my current position, I have started many collaborations with local groups across fields (in research as well as in teaching), and would greatly enjoy exploring integration potential with EPFL’s research teams in Computer Science and Robotics, as well as with groups outside of the exact and technical sciences.

With respect to my core research goal of ideal interfaces in hybrid socio-technical systems, I expect that the ecological and affordance-based approach that underlies my research should resonate with several of EPFL’s groups efforts in designing intuitive, human-centered interaction paradigms that merge physical and digital worlds and that increasingly (need to) integrate AI agents as well. A research integration should be particularly fruitful with EPFL’s *CHILI Lab*—while with *ideal interfaces* I share the same goal of contributing to more efficient (collaborative) problem-solving, I am in particularly interested in their framing of “signal-level computing in collaboration technologies”, which resonates very much with my affordance-oriented perspective as well as the technologies used in my lab (in particular, gaze). My research in the area of *Personalized Reality* would integrate very well with the immersion-oriented full-body interfaces of EPFL’s *Immersive Interaction Research Group* and, of course, with the EPFL *HCI Group* where health effects and responsible usage of AI systems seem to be common interests. My research furthermore has a lot of correspondence with the EPFL *REACT Group*’s stated research goals and publications (albeit I do not focus on learning science). Finally, I would explore integration potential with EPFL’s *VILAB*, where my research brings many relevant Ubiquitous Computing application scenarios—from industry to health to social media—that benefit from state-of-the-art visual intelligence.

With respect to robotics, my research builds on principles of interoperability and semantic reasoning that I believe are directly relevant to EPFL’s robotics and autonomous systems initiatives, specifically work in the *Reconfigurable Robotics Lab*. In my current research, the domains of robotics, the industrial IoT, and smart buildings have yielded fruitful application use cases and, with our application partners, we have already applied our research to enable higher autonomy of and more intuitive interaction with interoperable industrial robots, autonomous vehicles, virtual support services, and building automation systems (e.g., [7, 11, 36, 38, 52, 61, 62, 64, 82]). I believe that, beyond the *School of Computer and Communication Sciences* and the *School of Engineering*, groups in EPFL’s *School of Architecture, Civil and Environmental Engineering* might be interested in collaboration on these topics—and I would enjoy taking inspiration from groups such as ENAC’s *Lab for Creative Computation*. To support a flow of research results to industrial application, I uphold (partially formal) partnerships with industrial companies such as Siemens AG, and have recently contributed as a founding member to the W3C Community Group on *Autonomous Agents on the Web*⁹ to strengthen standardization efforts, which have in the past—with the W3C WoT—been highly conducive to bring research to practice.

Interdisciplinary Research

I very much enjoy deep interdisciplinary work and would like to continue this as well as part of a new position across three main fields and, potentially, with new partners at EPFL.

Nutrition and Health. At EPFL, I would continue research that I am currently conducting in the health and nutrition field together with medical doctors and partners across Switzerland, Germany, and Austria (e.g., [1, 44, 56, 60, 83, 84]). In this field, one of my current Ph.D. students is currently attempting to spin off our

⁹See <https://www.w3.org/community/webagents/>

nutrition-oriented research in the start-up company NutriBuddy¹⁰, and I would like to continue this research and combine it with the work of EPFL's Prof. Marcel Salathé, specifically since our respective approaches to nutrition tracking seem very synergetic. I have furthermore recently entered into a collaboration with doctors in the field of neuropsychiatry where we apply Ubiquitous Computing and AI methods for median nerve 3D reconstruction from ultrasound imaging, an application domain that I find highly stimulating and relevant and where I would like to follow up on our early research successes [18] and approach EPFL's *Biomedical Imaging Group*.

Regulation of Autonomous Systems. Originally driven by my research in the field of autonomous industrial systems, I have built significant expertise and partnerships with respect to the legal grounding, societal acceptance, and regulation of autonomous systems. While my group has been researching this topic for almost seven years (cf. [20, 24, 23, 25, 26, 22, 76, 78, 79, 80]) which is integrated in the recent book *AI and Law: How Automation is Changing the Law* [21], this research is gaining more and more societal relevance and momentum with the advancement of AI methods and their large-scale deployment in practice, and is currently spreading to further fields in Computer Science [77]. I would very much like to continue it in a new position at EPFL, again in the scope of collaborations with my current research partners, and would explore potential synergies with EPFL's *Center for Digital Trust* teams.

Systemic Risks and Online Platforms. Based on my successful long-term partnership with partners from legal scholarship and prompted by the high societal relevance of the topic, I have recently intensified our research of techno-legal methods to counter systemic risks from very large online platforms (VLOPs). We recently solidified our early results ([3]) and contributions to public media¹¹ in this line of research through the SNSF-funded project *CoCoDa* with the universities of Lausanne and Maastricht and the London-based Open Data Institute¹². While *CoCoDa* mainly targets improving the regulation of VLOPs on a global level, we also aim at local impact, for instance through the conceptualization of systems that protect vulnerable users, currently in a collaboration with a Swiss primary school¹³ that is funded by the Palatin Foundation.¹⁴ In a further recently awarded SNSF Bridge Discovery grant that is due to start in a few months, we are applying our findings to create an accessible, reliable, and collaborative AI system for better access to justice for underserved communities, with our partner *Caritas Switzerland*. I would continue these research projects in close collaboration with the University of Lausanne's Prof. Aurelia Tamò-Larrieux, who is, to my understanding, currently one of the individuals responsible for the interface between law and technology at EPFL.

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I was working on the above-mentioned research and practice projects over the past seven years with a dynamic team of postdoctoral researchers: Dr. Andrés Gómez (today Associate Professor at TU Braunschweig), Dr. Kimberly Garcia (accepted a Tenure-Track Assistant Professorship at ITU:T Linz), Dr. Andrei Ciortea (today an Assistant Professor at the University of St.Gallen), Dr. Clement Guitton (today a Senior Political Risk Manager at Swiss Re), and Dr. Kenan Bektas (today a Senior Researcher at Kantonsspital St.Gallen); and with several passionate Ph.D. students who graduated in the past year (Dr. Ganesh Ramanathan, Dr. Damian Hostettler, Dr. Sanjiv Jha), are awaiting their graduation ceremony (Ms. Danai Vachtsevanou), or are in earlier stages of their doctoral studies (Ms. Jing Wu, Mr. Jannis Strecker, Mr. Jérémy Lemée, Mr. Alessandro Giugno, Mr. Luka Bekavac, Mr. Jan Grau).

¹⁰See <https://www.nutribuddy.ch/>

¹¹See <https://www.zeit.de/digital/2025-02/rechts-tiktok-bundestagswahl-soziale-medien-afd>

¹²See <https://data.snf.ch/grants/grant/10004598>

¹³See <https://www.flag-safe.ch/>

¹⁴See <https://www.palatin.ch/projekt/flagsafe-deine-digitale-schutzstelle/>

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